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EchoPulse AI: A Hybrid Local Data Architecture for Multimodal Music Recommendation Integrating WASABI, MSD, and TalkPlay Datasets

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Abstract.
Music recommendation systems face a persistent challenge: emerging artists struggle to gain visibility because most algorithms prioritize popularity signals over content affinity, creating feedback loops that concentrate attention on established hits.
This paper presents EchoPulse AI, a locally-deployed multimodal recommender that addresses this gap through a Triple-Encoder Transformer architecture processing lyrical, acoustic, metadata, and conversational modalities.
The system consolidates heterogeneous datasets (WASABI, Million Song Dataset, Lyrics 1950-2019, and TalkPlayData-2) through a five-stage local pipeline without cloud dependencies.
Experimental evaluation against collaborative filtering, BERT4Rec, and SASRec baselines demonstrates that EchoPulse AI achieves competitive accuracy (\text{Hit}@10 = 0.312) while substantially improving Intra-List Diversity by 45.3% and Emerging Artist Discovery by 157.9%.
Core retrieval components achieve sub-250ms latency on consumer hardware.
These results suggest that content-aware multimodal fusion can mitigate popularity bias without sacrificing recommendation quality.
However, the current implementation is validated on a curated dataset of 5,710 tracks; scaling to industrial-scale catalogs and conducting online user studies remain important directions for future work.

Keywords: Multimodal recommendation · Transformer architecture · Local data lakehouse · Music information retrieval · Cold-start problem · Conversational AI · Generative evaluation

1 Introduction

The global music industry has experienced significant transformation driven by the transition to digital consumption models.
According to the International Federation of the Phonographic Industry (IFPI), global recorded music revenues reached USD 29.6 billion in 2024, representing continued growth in the streaming era [37].
However, this economic expansion has not translated proportionally into increased opportunities for emerging artists, who continue to face the fundamental challenge of visibility in an increasingly concentrated digital landscape.

Contemporary music recommendation systems predominantly rely on collaborative filtering approaches that leverage user-item interaction patterns.
While

effective for established content, these methods suffer from the well-documented cold-start problem and exhibit inherent popularity bias [24, 36].
New artists and niche content receive minimal exposure, creating feedback loops that concentrate audience attention on established hits.
This phenomenon is further exacerbated by the influx of algorithmically-generated content, which competes for limited recommendation slots.

To address these limitations, content-based approaches that leverage intrinsic song characteristics have gained research attention [35].
Multimodal systems that integrate lyrical, acoustic, metadata, and conversational features offer a pathway toward more equitable recommendation by prioritizing content affinity over historical popularity signals.
The Transformer architecture, originally introduced for natural language processing [1], has demonstrated remarkable effectiveness in capturing long-range dependencies across sequential data, making it particularly suitable for modeling the complex relationships inherent in musical content [40].

This paper presents EchoPulse AI, a locally-deployed multimodal music recommender system designed with three primary contributions: (1) a hybrid local data architecture following a five-stage paradigm that consolidates heterogeneous music datasets without cloud dependencies; (2) a Triple-Encoder Transformer architecture with specialized modules for lyrical, acoustic, conversational, and dialogue modalities, accompanied by rigorous mathematical formulations; and (3) an end-to-end evaluation framework demonstrating significant improvements in diversity and emerging artist discovery while maintaining competitive accuracy metrics, including novel metrics for evaluating generative transformer components.

The remainder of this paper is organized as follows.
Section 2 reviews related work on Transformer-based recommendation, multimodal music analysis, local data architectures, and conversational systems.

Section 3 details the methodology, including the five-stage data pipeline, dataset integration strategy, model architecture with mathematical formulations, and local inference pipeline.
Section 4 presents the experimental evaluation, covering hardware configuration, evaluation metrics, comparative performance analysis, diversity assessment, ablation studies, and latency measurements.
Section 5 discusses key findings, comparisons with existing architectures, limitations, and future research directions.
Section 6 concludes the paper with a synthesis of contributions and implications for equitable music recommendation.

2 Related Work

2.1 Transformer-Based Recommendation

The introduction of the Transformer architecture by Vaswani et al.
[1] revolutionized sequence modeling through the self-attention mechanism:

$$\operatorname{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \operatorname{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}} \right) \mathbf{V} \tag{1}$$

where Q, K, and V represent query, key, and value matrices respectively, and d_k denotes the dimensionality of key vectors.
This formulation enables models to capture long-range dependencies in sequential data, making it particularly suitable for modeling musical structure and user listening patterns.
BERT4Rec [4] introduced bidirectional encoding for sequential recommendation, improving accuracy but inheriting popularity bias from training data.
Similarly, SASRec [3] leveraged self-attention for sequential recommendation, establishing strong baselines for comparison.

2.2 Multimodal Music Recommendation

Multimodal approaches that integrate multiple data sources have shown promise for improving recommendation quality and diversity [40].
Research has explored fusion techniques for combining audio features with textual metadata [12], while cross-modal attention mechanisms have been investigated for music understanding [18, 19].
The WASABI dataset [6] and Million Song Dataset [5] provide foundational resources for multimodal music research, enabling large-scale analysis of lyrical, acoustic, and metadata features.
Recent work on music emotion recognition leverages the circumplex model of affect [42, 30] to map acoustic and lyrical features to emotional dimensions.

2.3 Local Data Architecture for ML Systems

The Lakehouse paradigm has emerged as a modern approach to data management, combining the flexibility and scalability of data lakes with the transactional capabilities of data warehouses [15].

While technologies such as Delta Lake, Apache Iceberg, and Apache Hudi enable ACID transactions over object storage, our implementation adapts these principles to a local environment using parquet-based storage with manual versioning, schema validation, and snapshot isolation—providing reproducibility and traceability without cloud infrastructure dependencies [45].

2.4 Conversational Recommendation Systems

Recent work such as TalkPlay [8] employs large language models for music recommendation through natural language interfaces.
However, these approaches often lack structured multimodal integration and rigorous evaluation of generative components.
Our work extends this direction by incorporating a dedicated conversational encoder trained on the TalkPlayData-2 dataset (\sim16,000 structured dialogues) and proposing evaluation metrics specific to intention-aware recommendation [26, 27].
Retrieval-augmented generation frameworks [16, 17] provide methodological foundations for integrating external knowledge sources with generative models.

EchoPulse AI Multimodal Music Recommender System Architecture

Fig.
1. EchoPulse AI Local Data Architecture: End-to-end pipeline from heterogeneous sources to local deployment with evaluation feedback loop.
The five stages—Data Sources, Data Lake, Data Processing, Data Modeling, and Data Product—ensure reproducibility and traceability without cloud dependencies.
Source: Own elaboration.

3 Methodology

3.1 Data Architecture Overview: Five-Stage Local Pipeline

The EchoPulse AI system implements a five-stage data pipeline following established MLOps principles, adapted for local deployment.
Figure 1 illustrates the end-to-end architecture from heterogeneous data sources to local deployment with an evaluation feedback loop.

3.2 Data Sources and Integration

The system integrates four primary data sources, each contributing distinct modalities:

WASABI Dataset [6]: Provides lyrical content and cultural metadata for over 2 million commercially released songs, with temporal coverage spanning 1922 to 2022.

Million Song Dataset (MSD) [5]: Offers 26 continuous acoustic features including tempo (\tau), energy (E), valence (V), danceability (D), and loudness (L) for 280,000 tracks, normalized to [0, 1] [33].

Lyrics 1950-2019

TalkPlayData-2 [8]: A novel conversational corpus containing ~16,000 structured dialogues mapping user intentions to song selections.
Each dialogue d_i is structured as:

di = \{(ut, rt, ct)\}{t=1}^{Ti} \to s^* \tag{2}

where ut denotes user utterance, rt system response, c_t contextual metadata, and s^* the target Spotify track ID.

The integration challenge lies in reconciling different identification schemes across datasets.
The MSD uses Echo Nest IDs, while WASABI maintains its internal identifier system.
Our solution employs heuristic matching based on the tuple (songname, artistname, release_year) with Levenshtein distance threshold \delta \le 2

High Human Impact High AI Impact

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.
See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.